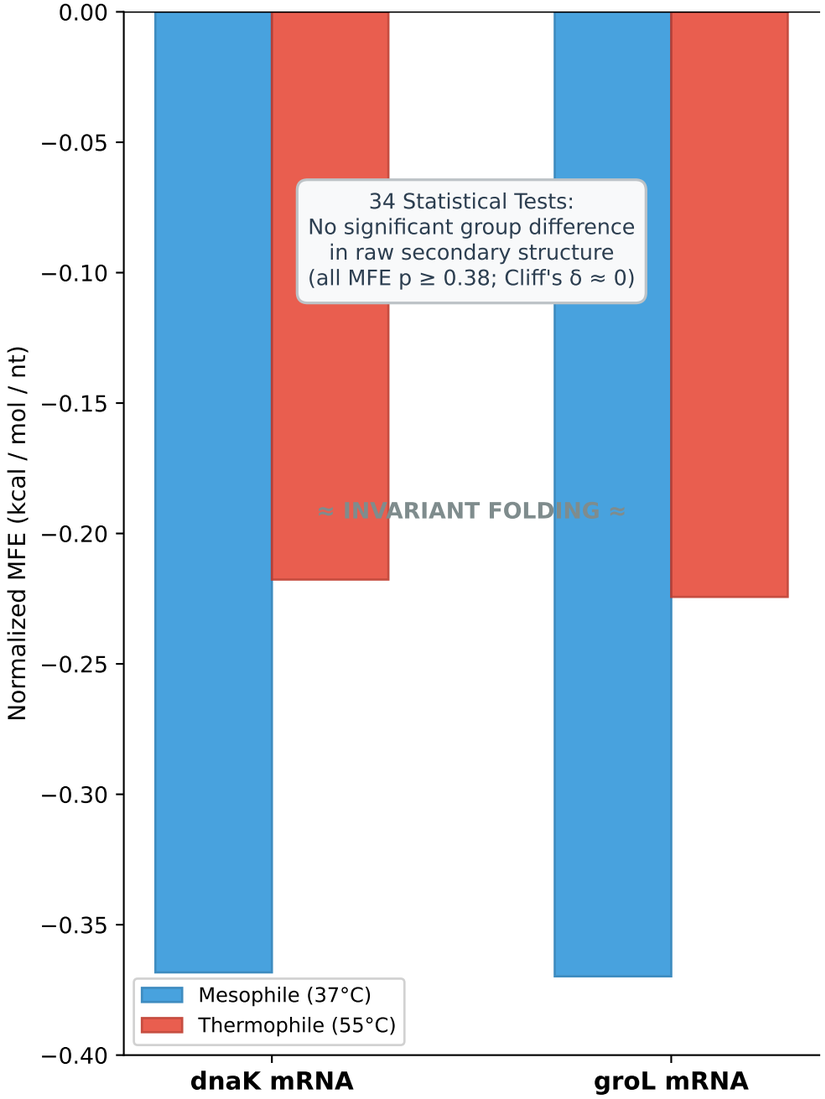
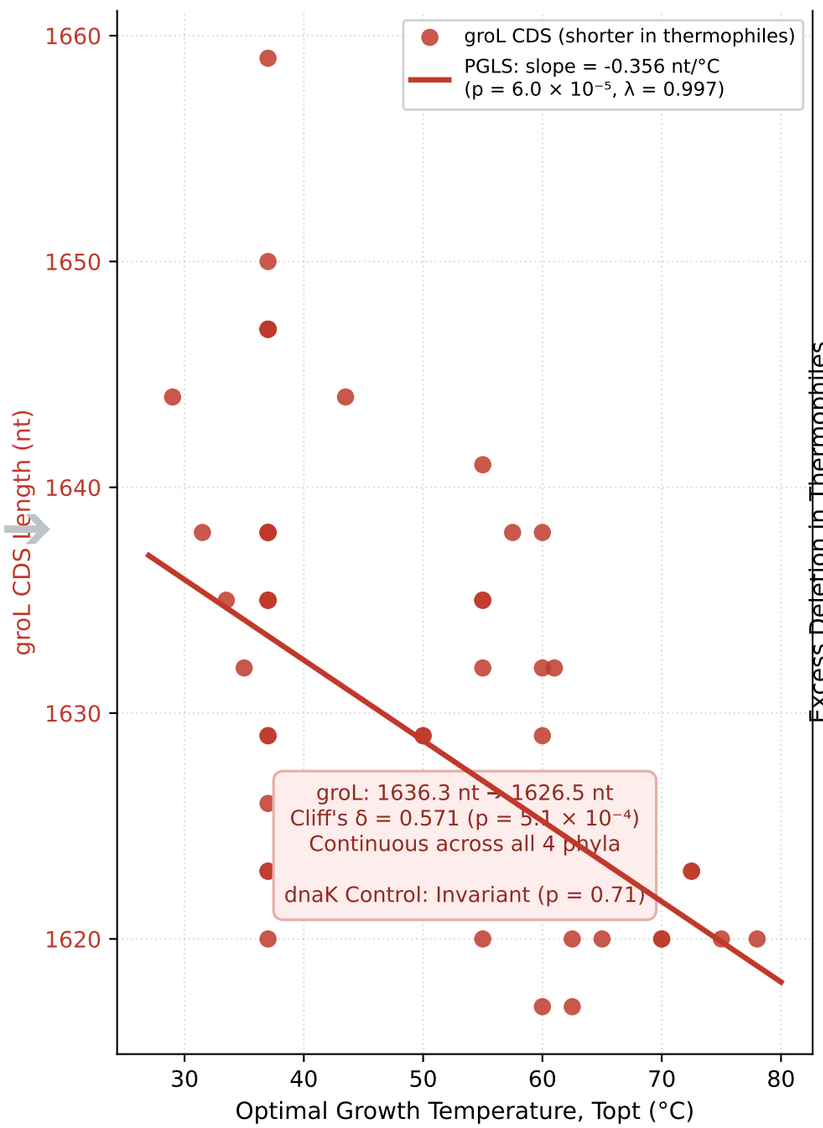
# Chaperone mRNA Secondary Structure & GroEL C-Terminal Tail Shortening

*Thermal Adaptation in Bacterial Chaperones Acts on Disordered Protein Extensions Rather Than Transcript Folding*

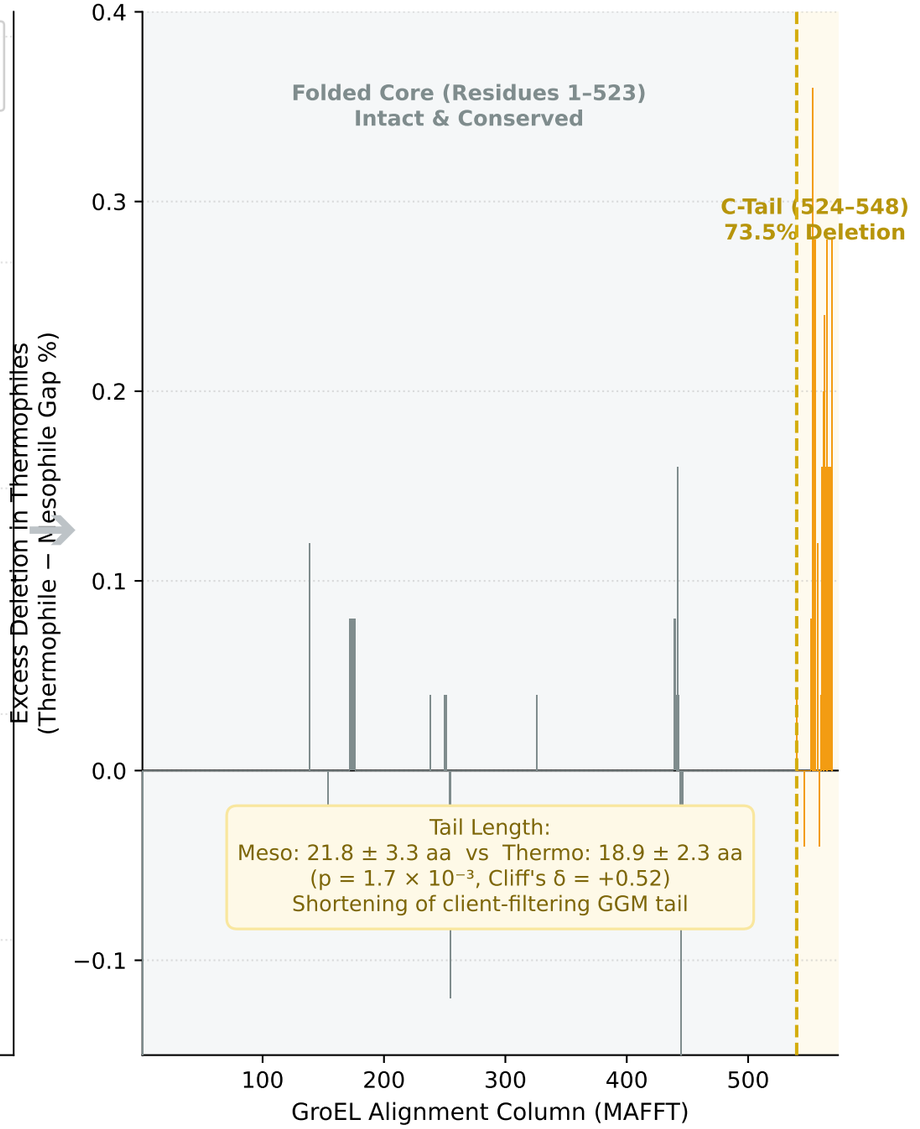
## 1. mRNA Secondary Structure




## 2. Coding Sequence Shortening




## 3. GroEL C-Terminal Tail Truncation




**Summary: Bacterial chaperone mRNA structure does not track thermal niche; instead, GroEL selectively shortens its disordered C-terminal peptide tail.**